

Dense Supervision, Sparse Updates: On the Sparsity and Geometry of On-Policy Distillation

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Abstract

On-policy distillation (OPD) has recently become a prominent post-training recipe by combining two desirable ingredients: on-policy student trajectories and dense teacher supervision. However, how this hybrid changes a model’s parameters remains unclear. Across several language and vision-language model pairs and OPD use cases, our analysis yields two main findings. On sparsity, OPD updates are small and coordinate-sparse. They are distributed across layers, with the largest relative movement usually appearing in FFN modules. This sparse structure is operationally useful: training only the discovered subnetwork nearly recovers full-training performance. The sparse support does not remove the need for adaptive optimization: SGD, previously reported to be competitive in RLVR, underperforms AdamW in our OPD optimizer ablation, suggesting that dense teacher supervision preserves useful momentum structure and heterogeneous second-moment scales. On geometry, the updates are numerically full-rank but spectrally concentrated; they lie mostly away from the principal singular subspaces of the source weights and fall disproportionately on coordinates where the source weights are close to zero. These findings suggest that dense teacher supervision does not turn OPD into ordinary dense parameter rewriting; instead, OPD retains important geometric signatures of on-policy post-training. Code is available at <https://github.com/SydCS/OPD-Param-Analysis>.

1 Introduction

On-policy distillation (OPD) is emerging as a third major component in post-training pipelines for large language models, alongside supervised fine-tuning and reinforcement learning (Yang et al., 2025; Li et al., 2026; Xiao et al., 2026; DeepSeek-AI, 2026). Recent practice uses OPD to transfer frontier-model reasoning into lightweight students, unify multiple post-training capabilities, or turn privileged information into self-improvement signals (Song & Zheng, 2026).

The reason for this momentum is conceptually intuitive. Conventional supervised fine-tuning and offline distillation provide dense token-level supervision, but train on fixed demonstrations that may differ from what the model will generate at test time, where distribution shift can cause compounding errors (Ross et al., 2011; Bengio et al., 2015). RLVR trains on the current policy’s own samples, but learns from environment- or verifier-derived outcome rewards that are often sparse, and therefore faces a credit-assignment problem. OPD, motivated by on-policy imitation approaches, combines the attractive parts of both: it keeps the data on-policy from the student model while replacing environmental reward feedback with dense teacher supervision (Agarwal et al., 2024). Table 1 summarizes this conceptual position relative to SFT, offline sequence-level distillation, and RLVR.

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Table 1: Conceptual position of OPD relative to SFT, offline SEQKD, and RLVR. OPD combines on-policy student samples with dense teacher feedback.

Training regime	Sample source	Learning signal	Analogy
SFT	Fixed data	Dense token labels	Behavioral Cloning in imitation learning
Offline SEQKD	Teacher demonstrations	Dense token labels	Behavioral Cloning in imitation learning
RLVR	Current policy	Sparse trajectory reward	Policy-based reinforcement learning
OPD	Current policy	Dense teacher feedback	Interactive imitation learning

Despite its empirical success, how OPD changes a model remains less understood. Prior post-training pipelines were often organized as SFT followed by RLVR, and recent work shows that these two stages can behave very differently in weight space. [Mukherjee et al. \(2026a\)](#) find that densely supervised off-policy fine-tuning produces much denser parameter updates than sparse-reward RLVR, which instead modifies a relatively small subnetwork. [Zhu et al. \(2025\)](#) further argue that RLVR moves away from the principal directions of the source weights. These findings leave a gap precisely where OPD sits: if the data are on-policy, but the learning signal comes from dense teacher feedback rather than sparse environment reward, should the update resemble supervised fine-tuning, RLVR, or constitute a distinct regime?

This paper answers this question by analyzing the sparsity and geometry of OPD parameter updates. We compute parameter updates between source models and OPD-trained models, measure their norm and coordinate sparsity, localize their layer and module structure, compare their overlap with RLVR updates, and analyze their singular spectra and alignment with source-weight singular subspaces. Results across several model pairs give a coherent picture: OPD-style updates are small in relative norm, sparse at checkpoint precision, usually show the largest relative movement in FFN modules, numerically full-rank yet spectrally concentrated, and strongly biased away from source-principal coordinates toward coordinates where the source weights are close to zero. OPD subnetwork masks also overlap with RLVR or OPD masks obtained with different teachers far above random baselines.

We further test whether these signatures matter operationally through two interventions. First, we restart OPD from the source model and train only coordinates selected by nonzero checkpoint-delta masks. The mask discovered from the full OPD run recovers nearly the same reasoning performance as full training, while density-matched random masks generally perform worse. Second, we compare AdamW and SGD under the same OPD setting and inspect AdamW optimizer states. SGD lags behind AdamW, suggesting that sparse final-update support does not by itself make adaptive coordinate-wise scaling unnecessary.

Together, these results show that dense teacher supervision does not make OPD an ordinary dense parameter-rewriting process. Although OPD lies between supervised fine-tuning and RLVR in objective design, its parameter updates are closer to sparse on-policy editing than to dense supervised rewriting. This points to the on-policy data distribution, not reward sparsity alone, as a major determinant of post-training update geometry.

2 Background

2.1 OPD as On-Policy Dense Supervision

OPD can be formulated as dense teacher supervision applied to on-policy student behavior ([Agarwal et al., 2024](#)). In ordinary supervised fine-tuning or offline knowledge distillation, the model is optimized on a fixed dataset of target sequences ([Hinton et al., 2015](#); [Kim & Rush, 2016](#)). In RLVR, the model samples from its current policy and receives an environment- or verifier-derived reward signal, often at the sequence level ([Shao et al., 2024](#)). In OPD, the student also samples from its current policy, but the teacher supplies token-level feedback on the sampled sequence. This gives OPD the data distribution of on-policy learning and the dense supervision of distillation.

Formally, let π_θ be the student policy and π_T be the teacher. For a prompt $x \sim \mathcal{D}$, OPD first samples a response $y = (y_1, \dots, y_{|y|}) \sim \pi_\theta(\cdot|x)$, and then minimizes a teacher-student divergence on the student-generated trajectory:

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x)} \left[\sum_{t=1}^{|y|} D(\pi_T(\cdot|x, y_{<t}) \parallel \pi_\theta(\cdot|x, y_{<t})) \right], \quad (1)$$

where D can be instantiated as forward / reverse KL divergence, JS divergence, or another token-level f -divergence. The key point is that the conditioning prefixes $y_{<t}$ come from the student, not from a fixed teacher trace.

This distinction turns OPD into a useful diagnostic for post-training dynamics. If sparse RLVR updates were mainly a consequence of sparse environment rewards, then replacing those rewards with dense teacher-derived feedback might be expected to produce much denser parameter changes. If instead the key ingredient is that the model trains near its own policy distribution, then OPD should retain part of the sparse and off-principal structure observed in RLVR, despite changing both the source and granularity of the learning signal.

2.2 Parameter Dynamics in LLM Post-Training

Recent work has begun to analyze post-training through parameter updates rather than only through reward curves or benchmark scores. Mukherjee et al. (2026a) show that RLVR often changes only a small subnetwork of an LLM, with subnetwork-only fine-tuning recovering the full run. Zhu et al. (2025) argue that this apparent sparsity reflects a model-conditioned geometry: RLVR updates are biased away from the principal directions of the source weights and toward low-curvature, spectrum-preserving regions. Mukherjee et al. (2026b) connect these dynamics to optimization, showing that SGD can be surprisingly competitive with AdamW in RLVR and can induce even sparser visible updates. These findings raise the question of what parameter characteristics OPD induces.

3 Analysis of Parameter Updates

We use model checkpoint deltas to ask three complementary questions about what OPD changes in weight space. First, scale and coordinate-support metrics ask whether the final update is globally small and whether it is written into many or few visible coordinates. Second, spectral metrics ask whether 2D weight-matrix updates are low-rank in a strict sense, or instead full-rank with concentrated singular-value energy. Third, source-geometry metrics ask whether updates reuse the dominant singular directions of the source weights or write into off-principal, low-magnitude coordinates.

We examine the parameter updates between source and trained checkpoints across ten model pairs for the scale, sparsity, spectral, and source-geometry analyses in Sections 4 and 5. See Appendix A.3 for details. Six are OPD-style pairs: two DS-Qwen¹ OPD runs distilled separately from the DeepScaleR and JustRL teachers (Luo et al., 2025; He et al., 2025), MiniCPM5-1B-SFT to MiniCPM5-1B OPD (OpenBMB, 2026), Qwen2.5-VL-3B-Instruct distilled from the NoisyRollout-7B teacher (Bai et al., 2025; Liu et al., 2026), Qwen3-1.7B-Base distilled from the Qwen3-4B-Base-GRPO teacher (Li et al., 2026), and Qwen3-4B to an OPSD checkpoint (Yang et al., 2025; Zhao et al., 2026). We also include three RLVR references: DeepScaleR-1.5B (Luo et al., 2025), JustRL (He et al., 2025), and a Qwen2.5-VL-3B-Instruct GRPO checkpoint. Finally, we include Qwen2.5-Math-1.5B to DS-Qwen as a conventional distillation-style contrast (Yang et al., 2024; Guo et al., 2025). Together, these analyzed model pairs cover both LLM and VLM settings, as well as the three application categories outlined in Appendix A.1: large-to-small distillation, capability consolidation, and self-distillation from privileged information.

For each matched floating-point tensor, we calculate the checkpoint delta by subtracting the stored weights loaded as bfloat16, $\Delta W = W_{\text{trained}} - W_{\text{src}}$, and use float32 arithmetic

¹We use DS-Qwen to denote DeepSeek-R1-Distill-Qwen-1.5B, a DeepSeek-R1 offline-distilled model in the Qwen2.5-Math family.

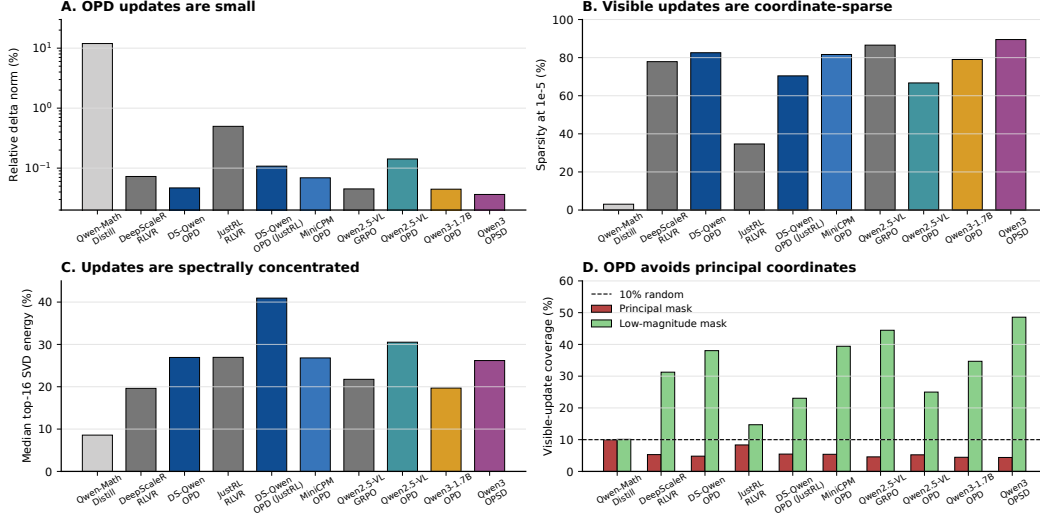


Figure 1: Checkpoint deltas show that OPD-style updates are small, coordinate-sparse, spectrally concentrated, and off-principal. Gray bars are reference runs rather than OPD runs: Qwen-Math Distill is an offline distillation-style contrast, and the remaining gray bars are RLVR references.

for norms, SVD, projections, and energy statistics. For scale and coordinate support, the relative delta norm $r = \|\Delta W\|_F / \|W_{\text{src}}\|_F$ measures update size relative to the source tensor, while visible coordinate sparsity $s_\epsilon = |\{i : |\Delta W_i| \leq \epsilon\}| / |\Delta W|$ measures the fraction of coordinates with no threshold-visible movement; we use $\epsilon = 10^{-5}$ in the main sparsity tables. We also measure coordinate energy concentration with $c_p = \sum_{i \in \text{Top}_p(|\Delta W|)} \Delta W_i^2 / \|\Delta W\|_F^2$, where Top_p denotes the largest $p\%$ coordinates by update magnitude. For spectral structure, we compute singular values $\sigma_i(\Delta W)$ for each 2D update matrix, top- k spectral energy $e_k = \sum_{i \leq k} \sigma_i^2 / \sum_i \sigma_i^2$, stable rank $\text{srnk}(\Delta W) = \|\Delta W\|_F^2 / \|\Delta W\|_2^2$, and numerical rank $\text{rank}_\tau(\Delta W) = |\{i : \sigma_i > \tau \sigma_1\}|$. Higher top- k energy and lower stable or numerical rank indicate stronger spectral concentration. For source-geometry alignment, the principal-subspace projection $p_k = \|U_k^\top \Delta W V_k\|_F^2 / \|\Delta W\|_F^2$, where $W_{\text{src}} = U \Sigma V^\top$, measures how much update energy lies in the leading source singular directions. The visible-update coverage $o(M) = \sum_i \mathbb{1}\{i \in M\} \mathbb{1}\{|\Delta W_i| > \alpha\} / \sum_i \mathbb{1}\{|\Delta W_i| > \alpha\}$ then measures how much of the visible update falls inside a source-principal coordinate mask or a low-magnitude source-weight mask. Full metric definitions and implementation details are given in Appendix A.4.

4 Sparsity of OPD Updates

4.1 Global Norm and Coordinate Sparsity

OPD-style updates are small relative to their source checkpoints. Across the six OPD-style pairs, the relative Frobenius norm ranges from 0.036% for Qwen3 OPSD to 0.142% for Qwen2.5-VL OPD. The visible coordinate sparsity is also high: at the 10^{-5} threshold, 66.72% to 89.50% of coordinates have no visible movement. The offline distillation contrast behaves very differently, with 11.94% relative delta norm and only 3.06% sparsity at the same threshold.

The key takeaway from these numbers is not that every OPD run has the same sparsity level, but that OPD does not become dense simply because of the dense teacher signal. Qwen2.5-VL OPD has the broadest update among the OPD-style checkpoints, yet it still has nearly two orders of magnitude smaller relative norm than the offline distillation contrast

Table 2: Global delta statistics. Percentages are computed from final checkpoint differences. Sparsity is the percentage of coordinates whose update is numerically indistinguishable from zero at tolerance 10^{-5} . Top-16 SVD energy and stable rank summarize spectral concentration. Principal and low-magnitude coverage are visible-update coverage scores with 10% source-coordinate masks. Gray text marks non-OPD reference runs.

Pair	Rel. norm (%)	Sparsity (%)	Top-16 SVD energy (%)	Stable rank	Principal coverage (%)	Low-mag. coverage (%)
Qwen-Math Distill	11.936	3.06	8.57	82.74	9.95	10.10
DeepScaleR RLVR	0.073	77.89	19.62	16.07	5.29	31.26
DS-Qwen OPD	0.047	82.58	26.92	9.97	4.81	38.04
JustRL RLVR	0.497	34.68	26.94	12.35	8.34	14.70
DS-Qwen OPD (JustRL)	0.108	70.43	40.94	7.25	5.46	23.04
MiniCPM OPD	0.069	81.65	26.81	12.32	5.39	39.42
Qwen2.5-VL GRPO	0.045	86.56	21.75	16.57	4.58	44.46
Qwen2.5-VL OPD	0.142	66.72	30.52	12.58	5.23	24.99
Qwen3-1.7B OPD	0.045	79.03	19.69	20.31	4.45	34.69
Qwen3 OPSD	0.036	89.50	26.19	7.77	4.39	48.57

and preserves clear coordinate concentration. This suggests that on-policy training can still keep the update local in weight space.

JustRL provides a useful boundary case for interpreting the RLVR references. Unlike DeepScaleR and Qwen2.5-VL GRPO, the JustRL checkpoint has a larger relative delta norm (0.497%), much lower visible sparsity (34.68%), and weaker source-geometry avoidance: its 10% principal-mask coverage is 8.34%, close to the random-density baseline, while its low-magnitude coverage is only 14.70%. A plausible explanation is that the long, single-stage JustRL training process accumulates a broader displacement more aligned with source-principal coordinates than shorter or more localized GRPO runs (He et al., 2025). Even so, JustRL remains spectrally concentrated, with 26.94% median top-16 SVD energy and stable rank 12.35, suggesting that densification in coordinate space does not erase the low-dimensional spectral profile. When the same JustRL checkpoint is used as the OPD teacher, the resulting OPD delta returns to the usual OPD-style pattern: small relative norm (0.108%), high sparsity (70.43%), strong spectral concentration, and off-principal source-geometry behavior. This boundary case suggests that the sparsity of the teacher’s own update is not a necessary condition for OPD to produce a sparse student update.

4.2 Layer and Module Structure

Having established that OPD updates are globally small and sparse, we next ask how this structure is distributed across layers and modules.

Figure 2 shows the DS-Qwen OPD sparsity profile by transformer layer and parameter matrix, indicating that OPD update sparsity is distributed across transformer layers rather than being isolated to a single layer. The average sparsity decreases from the low-80% range in early layers to around 70% in later layers, but all major projection matrices remain substantially sparse throughout the network. Among individual matrices, o_{proj} stays relatively sparse in later layers, while k_{proj} becomes visibly less sparse near the end of the model. This pattern suggests that OPD does not simply update a single layer block; it writes a sparse subnetwork across many layers and modules.

Table 3 reports the relative delta norm within each module group, $\|\Delta W_b\|_F / \|W_b\|_F$. This metric is the module-level analogue of the global relative norm in Table 2 and is less confounded by the number of parameters in each module. In most OPD-style runs, FFN modules have the largest or near-largest relative movement. However, the module pattern is not universal: MiniCPM OPD has comparable FFN, attention, and LM-head movement, Qwen3-1.7B OPD and Qwen3 OPSD have larger relative movement in attention, and the JustRL RLVR reference is dominated by the LM head. Normalization parameters remain nearly unchanged in all analyzed OPD checkpoints.

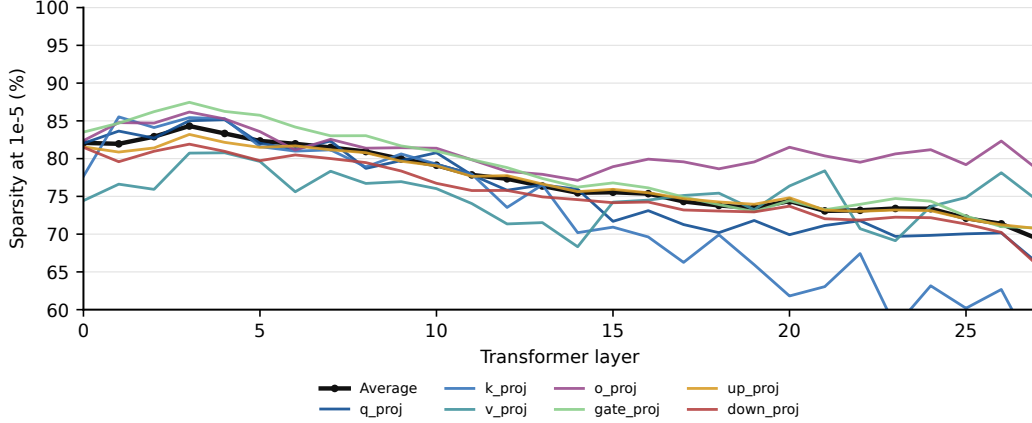


Figure 2: Layerwise and per-parameter-matrix update sparsity for DS-Qwen OPD. Sparsity counts coordinates whose delta between stored checkpoint weights is numerically indistinguishable from zero at tolerance 10^{-5} .

Table 3: Block-level relative delta norm, $\|\Delta W_b\|_F / \|W_b\|_F$, reported as a percentage. A dash in the LM-head column indicates that no matched standalone lm.head delta is available, because the output LM head is tied to the token embedding. The vision-tower column applies to the VLM checkpoint.

Pair	FFN	Attention	Embedding	LM head	Normalization	Vision tower
DeepScaleR RLVR	0.1136	0.0326	0.0512	0.0594	0.0009	–
DS-Qwen OPD	0.0741	0.0207	0.0171	0.0370	0.0002	–
JustRL RLVR	0.4643	0.3273	0.3365	1.6522	0.2693	–
DS-Qwen OPD (JustRL)	0.1720	0.0487	0.0277	0.0586	0.0008	–
MiniCPM OPD	0.0763	0.0675	0.0425	0.0689	0.0000	–
Qwen2.5-VL OPD	0.1629	0.1085	0.1121	–	0.0002	0.1580
Qwen3-1.7B OPD	0.0819	0.0934	0.0331	–	0.0000	–
Qwen3 OPSD	0.0376	0.0504	0.0078	–	0.0001	–

Sparsity and relative norm give complementary views of the same checkpoint delta. The sparsity profile counts how many coordinates in each layer or matrix remain unchanged under the numerical threshold, whereas the module relative norm measures how strongly each module group changes relative to its own source-weight scale. In the present checkpoints, the two views together show that OPD writes sparse updates broadly across layers, while the modules with the largest relative movement vary across models. This module structure suggests that future parameter-efficient OPD designs may need module-aware capacity allocation rather than a uniform budget.

4.3 OPD-RLVR Mask Overlap

To test whether OPD reuses the same sparse subnetwork as RLVR, we compare visible-update masks from checkpoint deltas using one-sided coverage because the mask densities can differ substantially. See Appendix A.5 for metric definitions.

Table 4 shows that the observed coverage is roughly three times the corresponding independent-mask baseline. On DS-Qwen, for example, the DeepScaleR RLVR and DS-Qwen OPD masks cover each other far above random expectation: 53.21% versus 17.50% in the RLVR-to-OPD direction, and 67.20% versus 22.11% in the reverse direction. The same pattern holds in the VLM setting, where the Qwen2.5-VL GRPO mask is much smaller than the OPD mask but 73.53% of its updated coordinates are covered by OPD, compared with a 33.28% baseline. It also holds across DS-Qwen OPD runs distilled from different teachers,

Table 4: Overlap between visible-update subnetworks. o_1 and o_2 are one-sided coverage scores; the Random column gives the corresponding independent-mask baseline for that one-sided score.

Model family	Setting	Random	Observed	Sparsity
DS-Qwen	$\mathcal{I}_1$: DeepScaleR RLVR	o_1 : 17.50	o_1 : 53.21 (3.04 $\times$)	77.89
DS-Qwen	$\mathcal{I}_2$: OPD (DeepScaleR teacher)	o_2 : 22.11	o_2 : 67.20 (3.04 $\times$)	82.58
Qwen2.5-VL	$\mathcal{I}_1$: GRPO	o_1 : 33.28	o_1 : 73.53 (2.21 $\times$)	86.56
Qwen2.5-VL	$\mathcal{I}_2$: OPD	o_2 : 13.44	o_2 : 29.69 (2.21 $\times$)	66.72
DS-Qwen	$\mathcal{I}_1$: OPD (DeepScaleR teacher)	o_1 : 29.57	o_1 : 80.17 (2.71 $\times$)	82.58
DS-Qwen	$\mathcal{I}_2$: OPD (JustRL teacher)	o_2 : 17.50	o_2 : 47.45 (2.71 $\times$)	70.43

DeepScaleR and JustRL (Luo et al., 2025; He et al., 2025), where coverage reaches 80.17% in one direction and 47.45% in the other.

These overlaps suggest that OPD, RLVR, and teacher-varied OPD do not merely produce sparse updates of the same density; they preferentially touch intersecting coordinates that may be related to a model’s reasoning capability.

4.4 Fine-Tuning the Subnetwork Alone Suffices in OPD

The first functional test asks whether the sparse OPD subnetwork is only a post-hoc description or whether it is sufficient for training. Given a source checkpoint θ_0 and a full OPD checkpoint θ_{full} , we define the full-OPD mask

$$m_i = \mathbb{1}\{|\theta_{\text{full},i} - \theta_{0,i}| > \tau\}, \quad (2)$$

where $\tau = 10^{-5}$ after subtracting the stored checkpoint weights. We restart OPD from θ_0 , use the same teacher and training configuration, zero gradients outside a chosen mask, and restore frozen coordinates after each optimizer step so that decoupled weight decay cannot move them. We compare four runs: full OPD, OPD restricted to the full-OPD nonzero mask, OPD restricted to the DeepScaleR RLVR nonzero mask, and OPD restricted to a random mask with the same density as the full-OPD mask. The mask densities come directly from the complement of the sparsity measurements in Table 2: roughly 17.5% for the DS-Qwen OPD mask and random mask, and 22.1% for the DeepScaleR RLVR mask.

Figure 3(a) shows that the discovered subnetwork is sufficient for recovering reasoning performance under OPD, using the average validation accuracy over AIME24 and AIME25. Full OPD reaches a peak mean@16 accuracy of 35.52%, while the OPD-mask run reaches 35.10% and has a slightly higher late-training average over the last five evaluation points (34.73% versus 34.27%). The RLVR mask also trains successfully, peaking at 34.69%, which is consistent with the overlap results above: the RLVR subnetwork is related to the OPD subnetwork, but it is not the same mask. The random mask improves over the source model but remains weaker, peaking at 32.92% and showing a lower late-training average. Appendix A.9 shows the same subnetwork-masked pattern in the Qwen2.5-VL OPD setting. Thus, the sparse mask is not merely a budget constraint; the particular coordinates selected by post-training matter. This makes the OPD delta a sparse task vector in the sense of task arithmetic: the delta $\theta_{\text{full}} - \theta_0$ is a direction in weight space that implements a behavioral change (Ilharco et al., 2022). Unlike classical lottery tickets, which are uncovered by pruning dense networks (Frankle & Carbin, 2018), the OPD task vector is naturally sparse at checkpoint precision. Masked training on the support of this sparse task vector is enough to recover full-training performance, suggesting that the visible OPD support is not only descriptive but operationally useful.

4.5 Do We Need AdamW in OPD?

The second functional test asks whether OPD needs AdamW (Kingma & Ba, 2014; Loshchilov & Hutter, 2017), motivated by recent evidence that SGD can be surprisingly

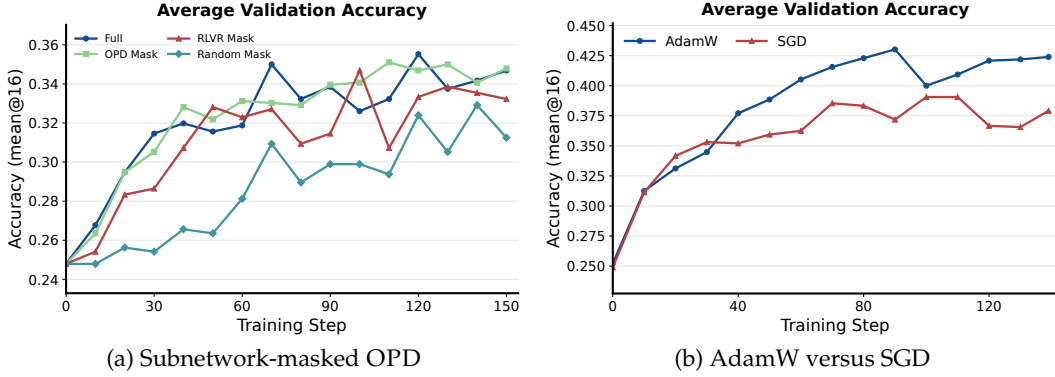


Figure 3: Comparison of OPD reasoning performance along the training process, reported as average validation accuracy over AIME24 and AIME25. Left: full OPD versus OPD trained with the OPD, RLVR, or random masks. Right: AdamW versus SGD in the JustRL-teacher OPD setting. Benchmark-wise curves are shown in Appendix A.7.

competitive for sparse RLVR updates (Mukherjee et al., 2026b). We run a direct optimizer ablation in the JustRL-teacher OPD setting: AdamW with learning rate 10^{-6} versus vanilla SGD with learning rate 10^{-2} . This follows prior optimizer analyses showing that AdamW’s small nominal rate can correspond to much larger effective per-parameter rates after second-moment normalization, so SGD is evaluated with a larger unnormalized learning rate rather than the same nominal value (Mukherjee et al., 2026b).

Figure 3(b) shows that SGD underperforms AdamW on the same reasoning validation suite, averaging mean@16 accuracy over AIME24 and AIME25. AdamW reaches a 43.02% peak and 42.40% final mean@16 accuracy, while SGD reaches a 39.06% peak and 37.92% final accuracy. Thus, the fact that OPD produces sparse final deltas is not sufficient to carry over the conclusion from RLVR that AdamW’s momentum and adaptive learning rates are largely unnecessary.

A likely reason is that OPD changes the learning signal, which in turn changes the optimization dynamics. Mukherjee et al. (2026b) argue for SGD in RLVR by showing that AdamW’s momentum can become poorly aligned with current gradients and that adaptive second-moment scaling is less load-bearing in their RLVR setting. For OPD, the distillation objective can produce a different gradient geometry: every generated token receives supervision from the same teacher rather than only a sparse and noisy reward signal. Appendix A.8 provides supporting diagnostics for this interpretation in the JustRL-teacher AdamW run. Momentum alignment is relatively high but unstable: it starts at 0.958, remains positive on average (0.280), drops to 0.008 by the end, and briefly becomes negative. This suggests that AdamW’s first moment is not simply uninformative in OPD, even though its fluctuating alignment makes momentum an incomplete explanation by itself. More importantly, the second-moment coefficient of variation remains large, averaging 4.85, suggesting that AdamW’s coordinate-wise scaling is still useful for OPD. In short, OPD shares sparse final-update structure with on-policy post-training, but its dense teacher signal can preserve enough gradient heterogeneity for AdamW to matter.

5 Geometry of OPD Updates

5.1 Full-Rank but Spectrally Concentrated

OPD updates are not low-rank in the strict numerical sense. Figure 1 and Table 2 summarize the main spectral evidence, with full spectral statistics in Appendix Table 9. For the six OPD-style pairs, the median numerical rank percentage is 100% under the relative singular-value threshold $\sigma_i > 10^{-5}\sigma_1$. However, this full-rank property coexists with spectral concentration. Their median top-16 singular-value energy ratio ranges from 19.69% to 40.94%. The corresponding median stable ranks range from 7.25 to 20.31. The offline distillation contrast

is also numerically full-rank, but its spectrum is much more diffuse: Qwen-Math Distill has only 8.57% top-16 energy and stable rank 82.74. By contrast, DeepScaleR RLVR is closer to OPD (19.62% top-16 energy and stable rank 16.07), suggesting that spectral concentration is more characteristic of sparse on-policy post-training than of offline distillation.

This distinction matters for interpreting low-rank adaptation. If one only reports numerical rank, OPD appears incompatible with low-rank parameterization. If one only reports top- k energy, OPD may appear effectively low-dimensional. A more accurate statement lies between these extremes: OPD deltas are full-rank matrices with a spectrally concentrated energy profile. Low-rank methods may therefore capture a meaningful part of the update energy, but they should not be expected to exactly reproduce the full checkpoint delta.

5.2 Off-Principal Movement from Source Weights

OPD updates have weak alignment with the principal singular directions of the source weights. For each source matrix $W_{\text{src}} = U\Sigma V^\top$, we measure

$$\frac{\|U_k^\top \Delta W V_k\|_F^2}{\|\Delta W\|_F^2}, \quad (3)$$

where U_k and V_k are the leading k left and right singular vectors of W_{src} . The main coverage scores are reported in Table 2, and the 1%, 5%, 10%, and 20% source-mask variants are given in Appendix Table 10. At a rank fraction of 10%, the median joint principal projection is below 1% for every main OPD-style pair. This means that OPD is not primarily amplifying the high-energy directions already dominant in the source weights. The offline distillation-style contrast shows a weaker version of this effect: Qwen-Math Distill has 2.22% median principal-projection energy, several times larger than DS-Qwen OPD and DeepScaleR RLVR.

The coordinate-mask analysis gives the same conclusion in a more local form. At the 10% mask setting, source-principal coordinates cover only 4.39% to 5.46% of visible OPD update coordinates, roughly half of the 10% random-density baseline. By contrast, the bottom 10% source-magnitude coordinates cover 24.99% to 48.57% of visible OPD update coordinates. The distillation-style contrast is close to random on both masks, with 9.95% principal coverage and 10.10% low-magnitude coverage. DeepScaleR RLVR shows the same qualitative bias as OPD, with 5.29% principal and 31.26% low-magnitude coverage. This contrast strengthens the interpretation that off-principal, low-magnitude writing is a signature of on-policy training; it also marks where OPD differs from offline distillation.

6 Discussion and Future Work

The primary message of this paper is that OPD produces small, sparse, spectrally concentrated, off-principal updates. Rather than interpolating linearly between SFT and RLVR in parameter space, OPD remains closer to sparse on-policy post-training. A qualitative explanation is that OPD trains the model near its own behavioral distribution. Unlike offline distillation, which may force the student to fit external trajectories far from what it currently generates, OPD provides teacher feedback on the student’s own rollouts. The student can therefore refine behavior in a familiar region of its policy space rather than broadly rewriting parameters to imitate off-policy traces. This offers one reason why dense token-level supervision can still produce such updates.

Our experiments mainly study OPD through static final-checkpoint analysis. However, we do not trace the full OPD training trajectory, so the learning dynamics that produce the final task vector remain to be revealed. More broadly, this paper focuses on how OPD changes a model in weight space; complementary behavioral analyses of how on-policy training changes model outputs remain an important direction (Shenfeld et al., 2025).

Another future direction concerns scale and coverage. Our observational experiments include relatively small-scale LLM and VLM checkpoints; interventional experiments are limited to DS-Qwen and Qwen2.5-VL for math reasoning. Larger models, additional domains such as agentic or embodied tasks, and more OPD applications are needed to

determine whether the observed sparsity, module structure, off-principal movement, and optimizer sensitivity are stable properties of OPD or artifacts of the present training recipe.

A third direction is OPD-native adaptation and optimization. The spectral concentration of OPD updates naturally suggests low-rank adaptation methods such as LoRA and its variants (Hu et al., 2021; Schulman & Lab, 2025), while the off-principal geometry suggests orthogonal fine-tuning methods (Liu et al., 2024). On the optimizer side, OPD’s matrix-level geometry may be incompatible with the spectral normalization introduced in vanilla Muon, making recent high-pass variants worth testing (Jordan et al., 2024; Liu et al., 2025; Fan et al., 2026). More broadly, these results point toward OPD-native fine-tuning methods and optimizers that allocate capacity according to the sparse, spectrally concentrated, off-principal task vector induced by on-policy distillation.

7 Conclusion

This work studies OPD parameter updates through two complementary properties: sparsity and geometry. Observational studies show that OPD-style updates are small, coordinate-sparse, spectrally concentrated but numerically full-rank, and strongly biased away from source-principal coordinates toward low-magnitude source coordinates. Interventional results show both the utility and the limits of this structure: training only the discovered OPD subnetwork is sufficient to recover full-OPD reasoning performance, but vanilla SGD underperforms AdamW in our optimizer ablation. Together, these findings place OPD closer to sparse on-policy editing than to dense supervised rewriting, even though its learning signal is dense and teacher-derived.

References

- Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos Garea, Matthieu Geist, and Olivier Bachem. On-policy distillation of language models: Learning from self-generated mistakes. In *International Conference on Learning Representations*, volume 2024, pp. 21246–21263, 2024.
- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-v1 technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. *Advances in neural information processing systems*, 28, 2015.
- Yuchen Cai, Ding Cao, Xin Xu, Zijun Yao, Yuqing Huang, Zhenyu Tan, Benyi Zhang, Guangzhong Sun, Guiquan Liu, and Junfeng Fang. On predictability of reinforcement learning dynamics for large language models. *arXiv preprint arXiv:2510.00553*, 2025.
- DeepSeek-AI. DeepSeek-V4: Towards highly efficient million-token context intelligence, 2026. URL <https://huggingface.co/deepseek-ai/DeepSeek-V4-Pro>.
- Chongyu Fan, Gaowen Liu, Mingyi Hong, Ramana Rao Kompella, and Sijia Liu. Rethinking muon beyond pretraining: Spectral failures and high-pass remedies for vla and rlvr. *arXiv preprint arXiv:2605.19282*, 2026.
- Jonathan Frankle and Michael Carbin. The lottery ticket hypothesis: Finding sparse, trainable neural networks. *arXiv preprint arXiv:1803.03635*, 2018.
- Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. Minillm: Knowledge distillation of large language models. In *International Conference on Learning Representations*, volume 2024, pp. 32694–32717, 2024.

- Etash Guha, Ryan Marten, Sedrick Keh, Negin Raoof, Georgios Smyrnis, Hritik Bansal, Marianna Nezhurina, Jean Mercat, Trung Vu, Zayne Sprague, et al. Openthoughts: Data recipes for reasoning models. *arXiv preprint arXiv:2506.04178*, 2025.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Bingxiang He, Zekai Qu, Zeyuan Liu, Yinghao Chen, Yuxin Zuo, Cheng Qian, Kaiyan Zhang, Weize Chen, Chaojun Xiao, Ganqu Cui, et al. Justrl: Scaling a 1.5 b llm with a simple rl recipe. *arXiv preprint arXiv:2512.16649*, 2025.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- Jonas Hübötter, Frederike Lübeck, Lejs Behric, Anton Baumann, Marco Bagatella, Daniel Marta, Ido Hakimi, Idan Shenfeld, Thomas Kleine Buening, Carlos Guestrin, et al. Reinforcement learning via self-distillation. *arXiv preprint arXiv:2601.20802*, 2026.
- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. *arXiv preprint arXiv:2212.04089*, 2022.
- Keller Jordan, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, and Jeremy Bernstein. Muon: An optimizer for hidden layers in neural networks, 2024. URL <https://kellerjordan.github.io/posts/muon/>.
- Yoon Kim and Alexander M Rush. Sequence-level knowledge distillation. In *Proceedings of the 2016 conference on empirical methods in natural language processing*, pp. 1317–1327, 2016.
- Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- Yaxuan Li, Yuxin Zuo, Bingxiang He, Jinqian Zhang, Chaojun Xiao, Cheng Qian, Tianyu Yu, Huan-ang Gao, Wenkai Yang, Zhiyuan Liu, et al. Rethinking on-policy distillation of large language models: Phenomenology, mechanism, and recipe. *arXiv preprint arXiv:2604.13016*, 2026.
- Jingyuan Liu, Jianlin Su, Xingcheng Yao, Zhejun Jiang, Guokun Lai, Yulun Du, Yidao Qin, Weixin Xu, Enzhe Lu, Junjie Yan, et al. Muon is scalable for llm training. *arXiv preprint arXiv:2502.16982*, 2025.
- Weiyang Liu, Zeju Qiu, Yao Feng, Yuliang Xiu, Yuxuan Xue, Longhui Yu, Haiwen Feng, Zhen Liu, Juyeon Heo, Songyou Peng, et al. Parameter-efficient orthogonal finetuning via butterfly factorization. In *International Conference on Learning Representations*, volume 2024, pp. 38317–38350, 2024.
- Xiangyan Liu, Jinjie Ni, Zijian Wu, Chao Du, Longxu Dou, Haonan Wang, Tianyu Pang, and Michael Shieh. Noisyrollout: Reinforcing visual reasoning with data augmentation. *Advances in Neural Information Processing Systems*, 38:2923–2957, 2026.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.
- Pan Lu, Ran Gong, Shibiao Jiang, Liang Qiu, Siyuan Huang, Xiaodan Liang, and Song-Chun Zhu. Inter-gps: Interpretable geometry problem solving with formal language and symbolic reasoning. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 6774–6786, 2021.

- Michael Luo, Sijun Tan, Justin Wong, Xiaoxiang Shi, William Y. Tang, Manan Roongta, Colin Cai, Jeffrey Luo, Li Erran Li, Raluca Ada Popa, and Ion Stoica. Deepscaler: Surpassing o1-preview with a 1.5b model by scaling rl. [Notion blog](#), 2025.
- Sagnik Mukherjee, Lifan Yuan, Dilek Hakkani-Tur, and Hao Peng. Reinforcement learning finetunes small subnetworks in large language models. *Advances in Neural Information Processing Systems*, 38:132119–132138, 2026a.
- Sagnik Mukherjee, Lifan Yuan, Pavan Jayasinha, Dilek Hakkani-Tür, and Hao Peng. Do we need adam? surprisingly strong and sparse reinforcement learning with sgd in llms. *arXiv preprint arXiv:2602.07729*, 2026b.
- OpenBMB. MiniCPM5-1B, 2026. URL <https://huggingface.co/openbmb/MiniCPM5-1B>.
- Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the fourteenth international conference on artificial intelligence and statistics*, pp. 627–635. JMLR Workshop and Conference Proceedings, 2011.
- John Schulman and Thinking Machines Lab. LoRA without regret. *Thinking Machines Lab: Connectionism*, 2025. doi: 10.64434/tml.20250929. <https://thinkingmachines.ai/blog/lora/>.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Idan Shenfeld, Jyothish Pari, and Pulkit Agrawal. RL’s razor: Why online reinforcement learning forgets less. *arXiv preprint arXiv:2509.04259*, 2025.
- Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. Hybridflow: A flexible and efficient rlhf framework. In *Proceedings of the Twentieth European Conference on Computer Systems*, pp. 1279–1297, 2025.
- Mingyang Song and Mao Zheng. A survey of on-policy distillation for large language models. *arXiv preprint arXiv:2604.00626*, 2026.
- Bangjun Xiao, Bingquan Xia, Bo Yang, Bofei Gao, Bowen Shen, Chen Zhang, Chenhong He, Chiheng Lou, Fuli Luo, Gang Wang, et al. Mimo-v2-flash technical report. *arXiv preprint arXiv:2601.02780*, 2026.
- An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, et al. Qwen2.5-math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- Zhuolin Yang, Zihan Liu, Yang Chen, Wenliang Dai, Boxin Wang, Sheng-Chieh Lin, Chankyu Lee, Yangyi Chen, Dongfu Jiang, Jiafan He, et al. Nemotron-cascade 2: Post-training llms with cascade rl and multi-domain on-policy distillation. *arXiv preprint arXiv:2603.19220*, 2026.
- Tianzhu Ye, Li Dong, Xun Wu, Shaohan Huang, and Furu Wei. On-policy context distillation for language models. *arXiv preprint arXiv:2602.12275*, 2026.
- Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tianfan Fan, Gaohong Liu, Lingjun Liu, et al. Dapo: An open-source llm reinforcement learning system at scale. *Advances in Neural Information Processing Systems*, 38:113222–113244, 2026.

Aohan Zeng, Xin Lv, Zhenyu Hou, Zhengxiao Du, Qinkai Zheng, Bin Chen, Da Yin, Chendi Ge, Chenghua Huang, Chengxing Xie, et al. Glm-5: from vibe coding to agentic engineering. *arXiv preprint arXiv:2602.15763*, 2026.

Siyan Zhao, Zhihui Xie, Mengchen Liu, Jing Huang, Guan Pang, Feiyu Chen, and Aditya Grover. Self-distilled reasoner: On-policy self-distillation for large language models. *arXiv preprint arXiv:2601.18734*, 2026.

Hanqing Zhu, Zhenyu Zhang, Hanxian Huang, DiJia Su, Zechun Liu, Jiawei Zhao, Igor Fedorov, Hamed Pirsiavash, Zhizhou Sha, Jinwon Lee, et al. The path not taken: RLvr provably learns off the principals. *arXiv preprint arXiv:2511.08567*, 2025.

A Appendix

A.1 Related Work

On-Policy Distillation. Early OPD formulations emphasize the exposure-mismatch problem in offline sequence-level distillation and train the student on its own sampled trajectories with dense teacher feedback (Agarwal et al., 2024; Gu et al., 2024). Recent work shows that this idea has become a practical post-training component for large language models (Song & Zheng, 2026), including large-to-small distillation (Yang et al., 2025; Li et al., 2026), capability consolidation with multiple teachers or post-training stages (Xiao et al., 2026; Zeng et al., 2026; Yang et al., 2026; DeepSeek-AI, 2026; OpenBMB, 2026), and on-policy self-distillation from privileged information (Zhao et al., 2026; Hübötter et al., 2026; Ye et al., 2026). Our work differs from these papers by studying the parameter update induced by OPD, rather than proposing a new OPD objective or training recipe.

Parameter Geometry of Post-Training. A second line of related work analyzes the parameter geometry of post-training. Mukherjee et al. (2026a) show that RLVR fine-tunes small subnetworks in large language models, while Zhu et al. (2025) argue that RLVR learns away from the principal directions of source weights. Cai et al. (2025) further study reinforcement-learning-induced parameter dynamics and find that dominant low-rank directions can predict later reasoning improvements. Mukherjee et al. (2026b) connect this sparse update regime to optimizer choice, showing that SGD can be surprisingly competitive for RLVR. We ask whether these properties persist when the sparse reward signal is replaced by dense teacher supervision, which places OPD between conventional dense-supervised fine-tuning and sparse-reward on-policy training. Our AdamW-versus-SGD result shows that the static sparsity analogy does not automatically carry over to optimizer choice.

Adaptation and Optimization Methods. Finally, the spectral and off-principal results relate to parameter-efficient adaptation and optimizer design. Low-rank adaptation methods such as LORA exploit the concentration of fine-tuning updates in a low-dimensional additive subspace (Hu et al., 2021; Schulman & Lab, 2025). Orthogonal fine-tuning methods such as OFT/BOFT instead parameterize adaptation through orthogonal transformations, aiming to preserve hyperspherical energy and pretrained representation structure while changing the model’s function (Liu et al., 2024). Muon is relevant from the optimizer side because it treats hidden-layer weights as matrices and orthogonalizes the momentum update before applying it (Jordan et al., 2024; Liu et al., 2025). Recent work argues that this spectral normalization can fail in RLVR post-training and proposes high-pass remedies, making these variants a more plausible starting point for OPD than vanilla Muon (Fan et al., 2026). These methods motivate future OPD-native optimization strategies.

A.2 Post-Training Objectives for LLMs

We summarize the training objectives that define the regimes compared in this paper. Let x denote a prompt, $y = (y_1, \dots, y_T)$ a response, π_θ the trainable policy, π_T a teacher policy, and π_{ref} a reference policy used for KL regularization when applicable.

SFT and Sequence-Level KD. Supervised fine-tuning (SFT) optimizes the negative log-likelihood of fixed target responses:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x, y^*) \sim \mathcal{D}} \sum_{t=1}^T \log \pi_\theta(y_t^* \mid x, y_{<t}^*). \quad (4)$$

Sequence-level knowledge distillation (SEQKD) uses the same maximum-likelihood objective, but the target response y^T is generated by a teacher rather than drawn from human annotations (Hinton et al., 2015; Kim & Rush, 2016):

$$\mathcal{L}_{\text{SeqKD}}(\theta) = -\mathbb{E}_{x \sim \mathcal{D}, y^T \sim \pi_T(\cdot \mid x)} \sum_{t=1}^T \log \pi_\theta(y_t^T \mid x, y_{<t}^T). \quad (5)$$

Both objectives are dense in the sense that every target token supplies a supervised learning signal, but the trajectories are offline-collected and fixed before the student update.

RLVR and GRPO. Reinforcement learning with verifiable rewards (RLVR) samples responses from the current or recent policy and optimizes a scalar reward $r(x, y)$, often with a KL penalty to keep the policy close to π_{ref} :

$$\mathcal{J}_{\text{RLVR}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x)} \left[r(x, y) - \beta \sum_{t=1}^T D_{\text{KL}}(\pi_\theta(\cdot|x, y_{<t}) \parallel \pi_{\text{ref}}(\cdot|x, y_{<t})) \right]. \quad (6)$$

GRPO is a seminal RLVR method that removes the learned critic by sampling a group of responses $\{y_i\}_{i=1}^G$ for the same prompt and normalizing their rewards within the group (Shao et al., 2024). With $\hat{A}_i = (r_i - \text{mean}(\{r_j\}_{j=1}^G)) / (\text{std}(\{r_j\}_{j=1}^G) + \epsilon)$ and $\rho_{i,t}(\theta) = \pi_\theta(y_{i,t}|x, y_{i,<t}) / \pi_{\text{old}}(y_{i,t}|x, y_{i,<t})$, a clipped GRPO objective can be written as

$$\begin{aligned} \mathcal{J}_{\text{GRPO}}(\theta) &= \mathbb{E}_{x, \{y_i\}_{i=1}^G} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} [g_{i,t}(\theta) - \beta k_{i,t}(\theta)] \right], \\ g_{i,t}(\theta) &= \min(\rho_{i,t}(\theta) \hat{A}_i, \text{clip}(\rho_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i), \\ k_{i,t}(\theta) &= D_{\text{KL}}(\pi_\theta(\cdot|x, y_{i,<t}) \parallel \pi_{\text{ref}}(\cdot|x, y_{i,<t})). \end{aligned} \quad (7)$$

RLVR therefore uses on-policy or near-on-policy samples but usually receives sparse sequence-level reward feedback.

OPD. OPD combines on-policy student rollouts with dense teacher feedback. One style, which we call GKD-style OPD, applies a token-level teacher-student divergence on prefixes generated by the student (Agarwal et al., 2024):

$$\mathcal{L}_{\text{OPD-GKD}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x)} \sum_{t=1}^{|y|} D(\pi_T(\cdot|x, y_{<t}) \parallel \pi_\theta(\cdot|x, y_{<t})), \quad (8)$$

where D is typically an f -divergence over token distributions. Another style, which we call PG-style OPD, treats distillation as policy optimization on student-generated samples (Gu et al., 2024). An abstract form is the reverse-KL sequence objective

$$\mathcal{L}_{\text{OPD-PG}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x)} \sum_{t=1}^{|y|} [\log \pi_\theta(y_t|x, y_{<t}) - \log \pi_T(y_t|x, y_{<t})], \quad (9)$$

which can be optimized with policy-gradient estimators because the sampled trajectory depends on the student policy. This PG view is connected to reverse-KL distillation through a simple score-function identity. For a fixed prefix $c = (x, y_{<t})$, define the sampled-token estimator $k_1(a; c) = \log \pi_\theta(a|c) - \log \pi_T(a|c)$. The full-vocabulary reverse KL at this prefix is

$$D_{\text{KL}}(\pi_\theta(\cdot|c) \parallel \pi_T(\cdot|c)) = \sum_a \pi_\theta(a|c) k_1(a; c). \quad (10)$$

Its gradient is

$$\begin{aligned} \nabla_\theta D_{\text{KL}}(\pi_\theta(\cdot|c) \parallel \pi_T(\cdot|c)) &= \nabla_\theta \sum_a \pi_\theta(a|c) k_1(a; c) \\ &= \sum_a [\nabla_\theta \pi_\theta(a|c)] k_1(a; c) + \sum_a \pi_\theta(a|c) \nabla_\theta k_1(a; c) \\ &= \mathbb{E}_{a \sim \pi_\theta(\cdot|c)} [k_1(a; c) \nabla_\theta \log \pi_\theta(a|c)] + \sum_a \nabla_\theta \pi_\theta(a|c) \\ &= \mathbb{E}_{a \sim \pi_\theta(\cdot|c)} [k_1(a; c) \nabla_\theta \log \pi_\theta(a|c)]. \end{aligned} \quad (11)$$

Here we use $\nabla_\theta \pi_\theta(a|c) = \pi_\theta(a|c) \nabla_\theta \log \pi_\theta(a|c)$ and $\nabla_\theta k_1(a; c) = \nabla_\theta \log \pi_\theta(a|c)$, since the prefix c and the teacher π_T are fixed. The final equality uses $\sum_a \pi_\theta(a|c) = 1$, so

$\sum_a \nabla_{\theta} \pi_{\theta}(a|c) = \nabla_{\theta} 1 = 0$. Thus, if k_1 is detached and $-k_1(a; c)$ is used as the policy-gradient advantage, the score-function loss $\mathcal{L}_{\text{PG}}(a; c) = \text{sg}[k_1(a; c)] \log \pi_{\theta}(a|c)$ has gradient $k_1(a; c) \nabla_{\theta} \log \pi_{\theta}(a|c)$, giving an unbiased single-sample estimator of the reverse-KL gradient up to practical modifications such as clipping, off-policy importance ratios, and advantage normalization.

Both OPD styles differ from SEQKD by training on student-generated trajectories, and differ from RLVR by replacing sparse scalar rewards with dense teacher-derived feedback.

A.3 Model Pairs and Training Details

Table 7 lists the checkpoint pairs used in the delta analysis. These pairs cover LLMs with PG-style OPD updates and VLMs with GKD-style OPD updates, and they span the three use cases of OPD discussed in Section A.1.

For training and evaluation, we use the public implementations and original configurations from ver1/HybridFlow (Sheng et al., 2025), the OPD codebase, and the OPSD codebase where applicable.² The DS-Qwen OPD runs use DAPO-Math-17K (Yu et al., 2026), the VLM GRPO and OPD runs use Geo3K (Lu et al., 2021), the Qwen3-1.7B OPD pair is a public checkpoint distilled from Qwen3-4B-Base-GRPO, and the OPSD run uses subsets of OpenThoughts_math_30k (Guha et al., 2025). Tables 5 and 6 summarize the core training configurations for the controlled DS-Qwen and Qwen2.5-VL OPD runs.

Table 5: Core training configuration for DS-Qwen OPD.

Setting	Value
Student model	DeepSeek-R1-Distill-Qwen-1.5B
Teacher model	DeepScaleR-1.5B or JustRL-DeepSeek-1.5B
Training / validation data	DAPO-Math-17K / AIME24 and AIME25
Distillation loss	k_1 policy-gradient style OPD; no task rewards
Rollout backend	vLLM rollout, $n = 4$ responses per prompt
Batching	train batch size 128; PPO mini-batch size 32
Sequence lengths	max prompt 1024; max train response 7168; max validation response 15360
Default optimizer	AdamW, learning rate 10^{-6} ; optimizer ablation uses SGD, learning rate 10^{-2}
Distributed setup	1 node, 4 actor GPUs per node, 2 teacher GPUs per node
Training schedule	1 epoch; evaluate every 10 steps

Table 6: Core training configuration for Qwen2.5-VL OPD.

Setting	Value
Student model	Qwen2.5-VL-3B-Instruct
Teacher model	NoisyRollout-Geo3K-7B
Training / validation data	Geo3K train / test split
Distillation loss	Forward-KL top- k OPD with $k = 32$; no task rewards
Rollout backend	vLLM rollout, $n = 1$ response per prompt
Batching	train batch size 128; PPO mini-batch size 32
Sequence lengths	max prompt 1024; max response 2048
Optimizer	AdamW, learning rate 10^{-6}
Distributed setup	1 node, 4 actor GPUs per node, 2 teacher GPUs per node
Training schedule	10 epochs; evaluate every 10 steps

A.4 Delta Metrics and Their Interpretation

This appendix gives the full metric definitions used in the checkpoint analysis introduced in Section 3. For each matched floating-point tensor with identical shape in the source and

²<https://github.com/ver1-project/ver1>, <https://github.com/thunlp/OPD>, and <https://github.com/siyan-zhao/OPSD>.

Table 7: Model pairs used in the delta analysis, spanning offline distillation, RLVR, OPD, and OPSD settings across LLM and VLM checkpoints, as well as large-to-small distillation, capability consolidation, and self-distillation use cases.

Pair	Source model	Trained model	Training regime / teacher or signal	Data
Qwen-Math Distill	Qwen2.5-Math-1.5B	DS-Qwen	Distillation reference; DeepSeek-R1 teacher	Public checkpoint
DeepScaleR RLVR	DS-Qwen	DeepScaleR-1.5B	RLVR; verifier/reward signal	Public RLVR checkpoint
DS-Qwen OPD	DS-Qwen	DS-Qwen OPD	OPD; DeepScaleR-1.5B teacher	DAPO-Math-17K
JustRL RLVR	DS-Qwen	JustRL	RLVR; simple fixed-hyperparameter recipe	Public RLVR checkpoint
DS-Qwen OPD (JustRL)	DS-Qwen	DS-Qwen OPD	OPD; JustRL teacher	DAPO-Math-17K
MiniCPM OPD	MiniCPM5-1B-SFT	MiniCPM5-1B OPD	OPD; MiniCPM5 RL+OPD recipe	Public OPD checkpoint
Qwen2.5-VL GRPO	Qwen2.5-VL-3B-Instruct	Qwen2.5-VL GRPO checkpoint	GRPO; verifier/reward signal	Geo3K
Qwen2.5-VL OPD	Qwen2.5-VL-3B-Instruct	Qwen2.5-VL OPD	OPD; NoisyRollout-7B teacher	Geo3K
Qwen3-1.7B OPD	Qwen3-1.7B-Base	Qwen3-1.7B-Base OPD	OPD; Qwen3-4B-Base-GRPO teacher	DAPO-Math-17K
Qwen3 OPSD	Qwen3-4B	Qwen3 OPSD checkpoint	OPSD; self-distillation with privileged traces	Subsets of OpenThoughts-math_30k

trained checkpoints, we form

$$\Delta W = W_{\text{trained}} - W_{\text{src}}. \quad (12)$$

The source-relative update scale is

$$r(W) = \frac{\|\Delta W\|_F}{\|W_{\text{src}}\|_F + \epsilon_0}, \quad (13)$$

where ϵ_0 is a small numerical stabilizer. This number measures how large the parameter displacement is relative to the original parameter energy; it does not by itself measure functional importance.

For a collection B of matched tensors, such as all analyzed tensors or all tensors assigned to one module group, we use the Frobenius-aggregated analogue

$$r(B) = \frac{\sqrt{\sum_{W \in B} \|\Delta W\|_F^2}}{\sqrt{\sum_{W \in B} \|W_{\text{src}}\|_F^2} + \epsilon_0}. \quad (14)$$

Table 2 reports this quantity with B equal to all matched tensors, while Table 3 reports it with B equal to one module group, such as FFN, attention, embedding, LM head, normalization, or the vision tower. The module-group value is therefore a relative delta norm within that group’s own source-weight scale. It is not the fraction of total update energy contributed by the group; a small module can have a high $r(B)$ while still accounting for little absolute update energy.

Coordinate sparsity measures the fraction of coordinates whose stored-checkpoint update is below a numerical threshold:

$$s_\epsilon(W) = \frac{|\{i : |\Delta W_i| \leq \epsilon\}|}{|\Delta W|}. \quad (15)$$

The implementation reports exact-zero sparsity, absolute-threshold sparsity for $\epsilon \in \{10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}\}$, and `isclose`($\Delta W, 0, \text{atol} = \epsilon$) sparsity for the same thresholds. The main paper reports `isclose` sparsity at $\epsilon = 10^{-5}$. We also compute relative-threshold sparsity,

$$s_\tau^{\text{rel}}(W) = \frac{|\{i : |\Delta W_i| \leq \tau \cdot \text{RMS}(W_{\text{src}})\}|}{|\Delta W|}, \quad \text{RMS}(W_{\text{src}}) = \sqrt{\frac{1}{|W|} \sum_i W_{\text{src},i}^2}, \quad (16)$$

which normalizes the threshold by the source tensor scale and makes the sparsity threshold less sensitive to the absolute scale of a particular tensor. Coordinate energy concentration is

$$c_p(W) = \frac{\sum_{i \in \text{Top}_p(|\Delta W|)} \Delta W_i^2}{\sum_i \Delta W_i^2 + \epsilon_0}, \quad (17)$$

where $\text{Top}_p(|\Delta W|)$ is the set of the largest $p\%$ coordinates by update magnitude. High c_p means that a small subset of coordinates carries a disproportionate amount of update energy.

For each 2D update matrix, we analyze the singular values of the update:

$$\Delta W = U_\Delta \Sigma_\Delta V_\Delta^\top, \quad \Sigma_\Delta = \text{diag}(\sigma_1, \dots, \sigma_m), \quad \sigma_1 \geq \dots \geq \sigma_m \geq 0. \quad (18)$$

The top- k spectral energy ratio is

$$e_k(W) = \frac{\sum_{i=1}^k \sigma_i^2}{\sum_{i=1}^m \sigma_i^2 + \epsilon_0}. \quad (19)$$

High e_k means that a rank- k approximation captures a large fraction of the update’s Frobenius energy. Stable rank is

$$\text{srank}(\Delta W) = \frac{\|\Delta W\|_F^2}{\|\Delta W\|_2^2 + \epsilon_0} = \frac{\sum_i \sigma_i^2}{\sigma_1^2 + \epsilon_0}. \quad (20)$$

Stable rank is a continuous proxy for rank based on energy spread: it is close to 1 when the top singular value dominates and increases as energy spreads across many singular directions. Numerical rank at tolerance τ is

$$\text{rank}_\tau(\Delta W) = |\{i : \sigma_i > \tau \sigma_1\}|. \quad (21)$$

Numerical rank counts how many singular values remain above a relative threshold; lower values indicate a more sharply truncated spectrum at that tolerance. Taken together, higher top- k energy and lower stable or numerical rank indicate stronger spectral concentration. Spectral concentration means that the update is directionally anisotropic and can be well approximated, in Frobenius energy, by a small number of rank-one singular components; it does not imply that the matrix is exactly low-rank or that the update is functionally unimportant. These quantities separate two notions that are often conflated: an update can be numerically full-rank while still having much of its energy concentrated in a small number of singular directions.

To measure alignment with the source-weight geometry, we decompose the source matrix as

$$W_{\text{src}} = U \Sigma V^\top. \quad (22)$$

For a rank fraction ρ , let $k = \max(1, \lfloor \rho \min(d_{\text{in}}, d_{\text{out}}) \rfloor)$, and let U_k, V_k be the leading source singular vectors. The joint principal-subspace projection is

$$p_k(W) = \frac{\|U_k^\top \Delta W V_k\|_F^2}{\|\Delta W\|_F^2 + \epsilon_0}. \quad (23)$$

Small p_k means that the update is not primarily written into the dominant singular directions of the source weight. The implementation also computes left-only and right-only source-principal projections, source-subspace rotation, and source spectral drift; the tables in this paper report the joint projection because it is the strictest alignment measure.

Finally, the coordinate-mask analysis converts source geometry into coordinate sets. The principal mask M_{prin} contains the coordinates with largest magnitude in the rank- k source reconstruction $U_k \Sigma_k V_k^\top$, while the low-magnitude mask M_{low} contains the coordinates with smallest $|W_{\text{src},i}|$. For an update mask $A_\alpha = \{i : |\Delta W_i| > \alpha\}$, mask coverage is

$$o_\alpha(M) = \frac{|A_\alpha \cap M|}{|A_\alpha| + \epsilon_0}. \quad (24)$$

In the static coordinate-mask tables produced by `analyze_opd_delta.py`, including the source-coordinate coverage scores reported in Table 2, the default is $\alpha = 0$, so the mask contains coordinates with exact nonzero checkpoint deltas. If M has density 10%, then 10% is the random-coverage baseline; values below 10% indicate avoidance, and values above 10% indicate enrichment. Low principal-mask coverage means the update avoids coordinates emphasized by the source’s leading singular structure, while high low-magnitude-mask coverage means the update preferentially occupies coordinates where the source weight was small in magnitude.

A.5 Visible-Update Mask Overlap Metrics

Table 4 compares two visible-update masks. For two checkpoint deltas $\Delta^{(1)}$ and $\Delta^{(2)}$, we first define their nonzero visible-update sets at threshold $\epsilon = 10^{-5}$:

$$A_1 = \{i : |\Delta_i^{(1)}| > \epsilon\}, \quad A_2 = \{i : |\Delta_i^{(2)}| > \epsilon\}. \quad (25)$$

The one-sided coverage scores are

$$o_1 = \frac{|A_1 \cap A_2|}{|A_1|}, \quad o_2 = \frac{|A_1 \cap A_2|}{|A_2|}. \quad (26)$$

These are asymmetric because each score normalizes by a different subnetwork size. For example, if A_1 is much smaller than A_2 , then A_1 can be mostly covered by A_2 even when only a modest fraction of A_2 is covered by A_1 .

The Random column reports the independent-mask baseline for the corresponding one-sided score. For o_1 , the random baseline is the density of the second mask,

$$\mathbb{E}_{\text{rand}}[o_1] = \frac{|A_2|}{N}, \quad (27)$$

where N is the total number of compared coordinates. Similarly, $\mathbb{E}_{\text{rand}}[o_2] = |A_1|/N$. This is why the two Random values for the same pair can differ: they are determined by the nonzero densities of the opposite masks.

A.6 Additional Static-Analysis Tables

Tables 8, 9, and 10 provide fuller static-analysis metrics for all analyzed model pairs. All percentages are computed from final checkpoint deltas. Exact zero and sparsity are computed after subtracting the loaded bfloat16 checkpoint weights, while relative-threshold sparsity uses the source tensor RMS as the scale. Spectral and source-geometry metrics are medians over analyzed 2D matrices.

Table 8: Additional global sparsity and coordinate-energy statistics for all analyzed model pairs. Sparsity@ ϵ uses $\text{isclose}(\Delta W, 0, \text{atol} = \epsilon)$. Rel.-thr. sparsity uses the threshold $10^{-3} \cdot \text{RMS}(W_{\text{src}})$.

Pair	Rel. norm (%)	Exact zero (%)	Sparsity@1e-6 (%)	Sparsity@1e-5 (%)	Sparsity@1e-4 (%)	Rel.-thr. sparsity (%)	Top 0.1% coord. energy (%)	Top 1% coord. energy (%)	Top 5% coord. energy (%)
Qwen-Math Distill	11.936	3.06	3.06	3.06	3.52	3.13	1.85	10.18	30.86
DeepScaleR RLVR	0.073	77.36	77.37	77.89	94.21	83.36	3.77	21.18	53.00
DS-Qwen OPD	0.047	82.02	82.04	82.58	98.35	89.34	2.73	19.28	55.41
JustRL RLVR	0.497	34.38	34.39	34.68	49.16	37.33	8.39	27.12	55.98
DS-Qwen OPD (JustRL)	0.108	69.96	69.99	70.43	86.16	74.24	1.66	13.42	39.57
MiniCPM OPD	0.069	80.46	80.49	81.65	98.91	89.05	2.80	22.93	58.05
Qwen2.5-VL GRPO	0.045	83.40	83.66	86.56	99.96	91.67	4.60	26.70	70.13
Qwen2.5-VL OPD	0.142	64.13	64.42	66.72	94.83	70.50	3.09	13.48	45.84
Qwen3-1.7B OPD	0.045	77.94	77.96	79.03	97.78	87.82	3.33	19.09	59.03
Qwen3 OPSD	0.036	86.50	86.67	89.50	100.00	94.52	3.95	27.99	67.21

Table 9: Additional spectral statistics for all analyzed model pairs. Top- k energy, stable rank, numerical rank, and source-principal projection are medians over analyzed 2D matrices.

Pair	Top-1 SVD energy (%)	Top-8 SVD energy (%)	Top-16 SVD energy (%)	Stable rank	Rank@1e-5 (%)	Principal proj. 10% energy (%)
Qwen-Math Distill	1.21	4.99	8.57	82.74	100	2.22
DeepScaleR RLVR	6.22	16.13	19.62	16.07	100	0.37
DS-Qwen OPD	10.03	22.74	26.92	9.97	100	0.37
JustRL RLVR	8.09	20.76	26.94	12.35	100	5.53
DS-Qwen OPD (JustRL)	13.78	34.32	40.94	7.25	100	0.53
MiniCPM OPD	8.11	21.50	26.81	12.32	100	0.38
Qwen2.5-VL GRPO	6.03	17.06	21.75	16.57	100	0.66
Qwen2.5-VL OPD	7.95	23.40	30.52	12.58	100	0.81
Qwen3-1.7B OPD	4.92	14.27	19.69	20.31	100	0.57
Qwen3 OPSD	12.87	23.11	26.19	7.77	100	0.47

Table 10: Source-geometry mask coverage for all analyzed model pairs. Principal coverage measures the fraction of visible-update coordinates falling in source-principal coordinate masks; low-magnitude coverage measures the fraction falling in the smallest-magnitude source coordinates. Non-prin./low measures the fraction of visible-update coordinates that either avoid the 10% source-principal mask or fall in the 10% low-magnitude source mask.

Pair	Prin. 1% (%)	Prin. 5% (%)	Prin. 10% (%)	Prin. 20% (%)	Low-mag. 1% (%)	Low-mag. 5% (%)	Low-mag. 10% (%)	Low-mag. 20% (%)	Non-prin./low 10% (%)
Qwen-Math Distill	1.00	4.98	9.95	19.90	1.01	5.05	10.10	20.18	90.53
DeepScaleR RLVR	0.58	2.82	5.29	10.15	3.46	16.56	31.26	55.17	96.16
DS-Qwen OPD	0.57	2.67	4.81	9.03	4.17	20.03	38.04	66.78	96.97
JustRL RLVR	0.86	4.24	8.34	16.57	1.52	7.49	14.70	28.25	92.34
DS-Qwen OPD (JustRL)	0.64	2.95	5.46	10.76	2.43	11.88	23.04	43.24	95.56
MiniCPM OPD	0.67	2.92	5.39	9.45	4.49	21.22	39.42	67.05	96.72
Qwen2.5-VL GRPO	0.58	2.55	4.58	7.83	5.35	24.83	44.46	71.91	97.30
Qwen2.5-VL OPD	0.64	2.87	5.23	9.63	2.69	13.03	24.99	45.84	95.80
Qwen3-1.7B OPD	0.53	2.44	4.45	7.98	3.86	18.41	34.69	60.79	97.09
Qwen3 OPSD	0.52	2.40	4.39	7.46	5.78	26.95	48.57	77.65	97.76

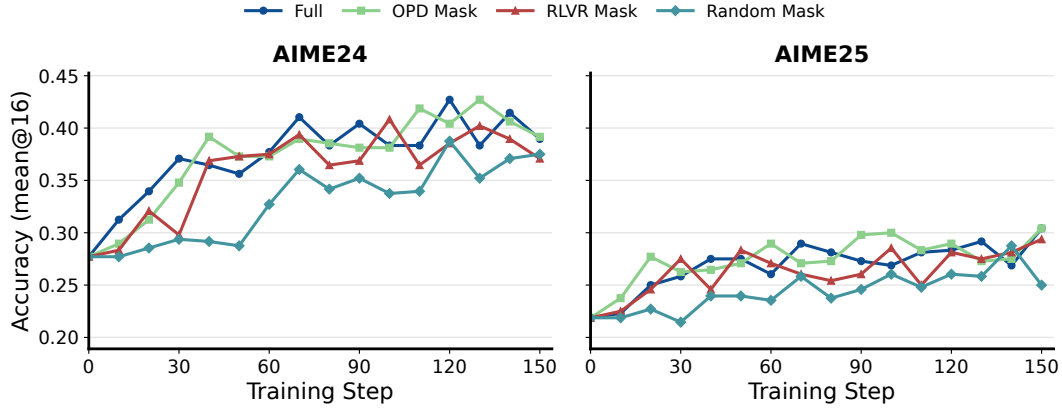


Figure 4: Benchmark-wise breakdown of the subnetwork-masked OPD experiment in Figure 3(a). Left: AIME24. Right: AIME25. The OPD mask tracks the full run closely on both benchmarks, while the random mask remains weaker.

A.7 Benchmark-Wise Breakdown of Interventional Experiments

The benchmark-wise curves support the same conclusions as the averaged curves in Figure 3. For the subnetwork intervention, the OPD mask closely tracks full training on both AIME24 and AIME25, while the random-mask run generally performs worse. For the optimizer ablation, SGD improves over the source checkpoint but trails AdamW on both benchmarks by the end of training, indicating that the sparse final-update structure does not by itself make AdamW unnecessary.

A.8 Optimizer Dynamics in OPD

We further inspect the AdamW optimizer state in the JustRL-teacher OPD run, following the same diagnostic perspective used by Mukherjee et al. (2026b). We use two optimizer-state metrics. First, momentum cosine similarity is computed as

$$\cos(m_t, g_t) = \frac{\langle m_t, g_t \rangle}{\|m_t\|_2 \|g_t\|_2},$$

where m_t is AdamW’s first-moment estimate and g_t is the current gradient after flattening the trainable parameters. Values close to 1 mean that the accumulated momentum points in the current descent direction, values near 0 mean that the first moment carries little directional agreement with the current gradient, and negative values mean that momentum is locally opposed to the current gradient. Second, the second-moment coefficient of variation is computed as

$$\text{CV}(\sqrt{v_t}) = \frac{\text{std}(\sqrt{v_t})}{\text{mean}(\sqrt{v_t})},$$

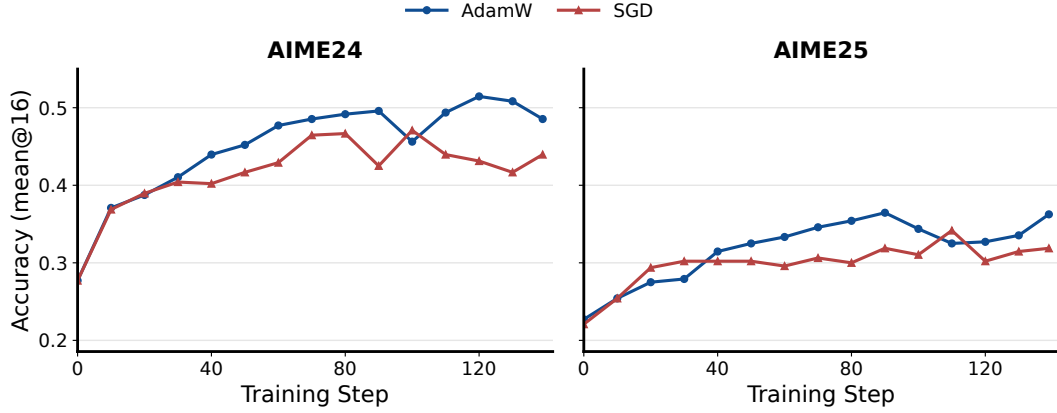


Figure 5: Benchmark-wise breakdown of the AdamW-versus-SGD experiment in Figure 3(b). Left: AIME24. Right: AIME25. In the JustRL-teacher OPD setting, SGD lags behind AdamW on both benchmarks at the final evaluation point.

where v_t is AdamW’s second-moment estimate. This metric measures how uneven AdamW’s adaptive denominator is across coordinates: a low value would make the adaptive update closer to uniform learning-rate rescaling, whereas a high value indicates substantial coordinate-wise variation in effective step sizes.

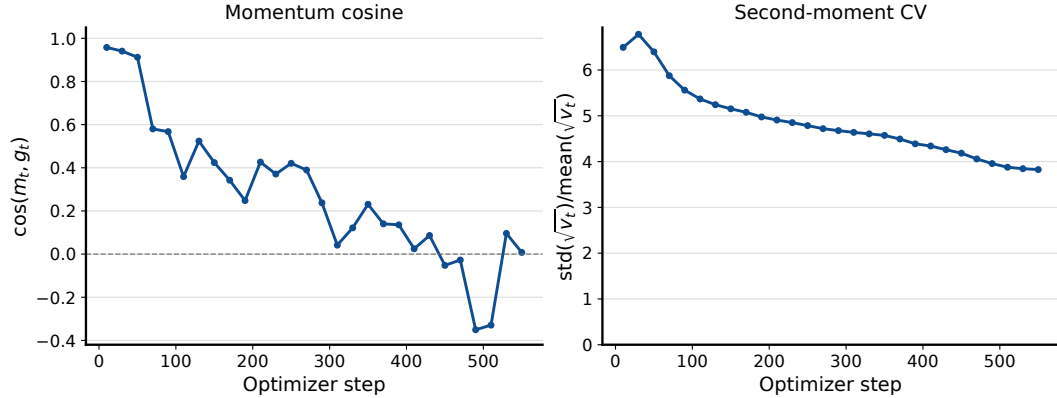


Figure 6: AdamW optimizer-state diagnostics for JustRL-teacher OPD. Left: cosine similarity between AdamW momentum and the current gradient. Right: coefficient of variation of $\sqrt{v_t}$, computed as $\text{std}(\sqrt{v_t})/\text{mean}(\sqrt{v_t})$. Momentum alignment is initially high but unstable, while the second-moment CV remains large throughout training.

Figure 6 shows that AdamW’s first and second moment statistics behave differently in this OPD run. The momentum cosine is relatively high but unstable: it starts at 0.958, remains positive on average with mean 0.280, falls toward zero, and reaches a minimum of -0.351 . This supports the view that momentum can retain meaningful directional history in OPD, but its alignment with the current gradient is not stable throughout training. By contrast, the second-moment CV remains high: it starts at 6.49, ends at 3.83, and averages 4.85. This persistent heterogeneity suggests that AdamW’s coordinate-wise adaptive scaling can still be load-bearing in OPD, even when the final checkpoint delta is sparse.

A.9 Qwen2.5-VL Subnetwork Experiments

We repeat the masked-subnetwork intervention in the Qwen2.5-VL OPD setting. All runs use the same Geo3K training configuration and are evaluated on the Geo3K validation set, reported as accuracy over the first 100 optimizer steps. We compare five variants: full

OPD, OPD restricted to the full-OPD nonzero mask, OPD restricted to the GRPO nonzero mask, OPD restricted to a random mask with the same density as the full-OPD mask, and OPD restricted to a random mask with the same density as the GRPO mask. The nonzero masks are again defined by $|\Delta| > 10^{-5}$; the full-OPD mask has roughly one-third active coordinates, while the GRPO mask has 9.3% active coordinates.

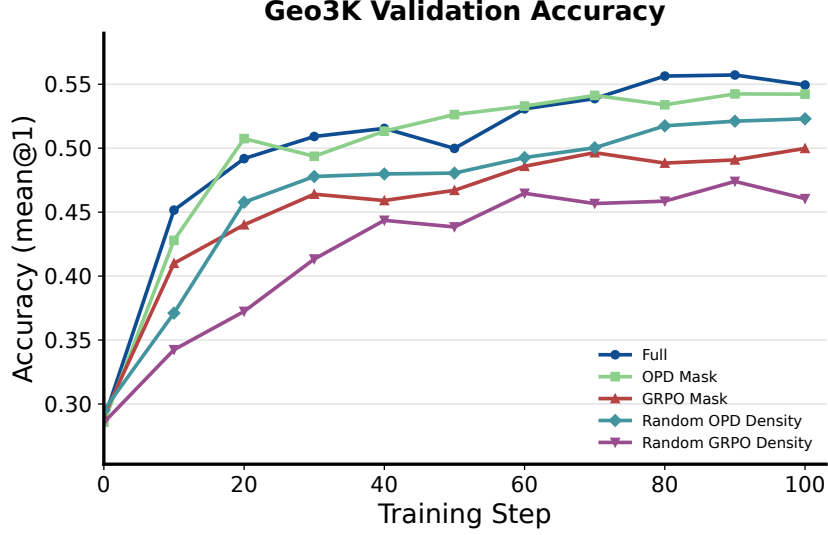


Figure 7: Qwen2.5-VL subnetwork-masked OPD on Geo3K. The full-OPD mask closely tracks the full run, while density-matched random masks and the smaller GRPO mask underperform.

Figure 7 shows the same qualitative pattern as the LLM subnetwork intervention. Full OPD reaches a peak Geo3K accuracy of 55.72% and a final-step accuracy of 54.94%, while the full-OPD-mask run reaches a 54.24% peak and 54.23% final accuracy. The late-training gap is similarly small: averaged over the last five evaluation points, full OPD obtains 54.65%, compared with 53.86% for the full-OPD mask. The random mask with the same density as the full-OPD mask peaks lower at 52.30%, showing that the full-OPD support is not merely a random one-third training budget. The GRPO mask reaches 49.98%, and the density-matched random GRPO mask peaks at 47.40%. Thus, in the VLM setting as well, the sparse coordinates selected by OPD are functionally useful for recovering most of the full-OPD improvement.